Definition 3: Circuit

Circuits are sub-graphs of the network, consisting of features and the weights connecting them.

language models remains challenging. However, there has been notable progress in scaling circuit analysis of *narrow behaviors* to larger circuits and models, such as indirect object identification (Wang et al., 2023) and greater-than computations (Hanna et al., 2023) in GPT-2 and multiple-choice question answering (Lieberum et al., 2023).

In search of general and universal circuits, researchers focus particularly on more general and transferable behaviors. McDougall et al. (2023)'s work on copy suppression in GPT-2's attention heads sheds light on model calibration and self-repair mechanisms. Davies et al. (2023) and Feng & Steinhardt (2023) focus on how large language models represent symbolic knowledge through variable binding and entity-attribute binding, respectively. Yu et al. (2023); Nanda et al. (2023c); Lv et al. (2024); Chughtai et al. (2024); Ortu et al. (2024) explore mechanisms for factual recall, revealing how circuits dynamically balance pre-trained knowledge with new contextual information. Lan & Barez (2023) extend circuit analysis to sequence continuation tasks, identifying shared computational structures across semantically related sequences.

More promisingly, some repeating patterns have shown *universality* across models and tasks. These universal patterns are called *motifs* (Olah et al., 2020) and can manifest not just as specific circuits or features but also as higher-level behaviors emerging from the interaction of multiple components. Examples include the curve detectors found across vision models (Cammarata et al., 2021; 2020), induction circuits enabling in-context learning (Olsson et al., 2022), and the phenomenon of branch specialization in neural networks (Voss et al., 2021). Motifs may also capture how models leverage tokens for working memory or parallelize computations in a divide-and-conquer fashion across representations. The significance of motifs lies in revealing the common structures, mechanisms, and strategies that naturally emerge across neural architectures, shedding light on the fundamental building blocks underlying their intelligence. Figure 6 contrasts observed neural network models with hypothetical disentangled models, illustrating how a shared circuit pattern can emerge across different models trained on similar tasks and data, hinting at an underlying *universality*.

Definition 4: Motif

Motifs are repeated patterns within a network, encompassing either features or circuits that emerge across different models and tasks.

Universality Hypothesis. Following the evidence for motifs, we can propose two versions for a *universality* hypothesis regarding the convergence of features and circuits across neural network models:

Hypothesis 3: Weak Universality

There are underlying principles governing how neural networks learn to solve certain tasks. Models will generally converge on analogous solutions that adhere to the common underlying principles. However, the specific *features* and *circuits* that implement these principles can vary across different models based on factors like hyperparameters, random seeds, and architectural choices.

The universality hypothesis posits a convergence in forming features and circuits across various models and tasks, which could significantly ease interpretability efforts in AI. It proposes that artificial and biological neural networks share similar features and circuits, suggesting a standard underlying structure (Chan et al., 2023; Sucholutsky et al., 2023; Kornblith et al., 2019). This idea posits that there is a fundamental basis in how neural networks, irrespective of their specific configurations, process and comprehend information. This could be due to inbuilt inductive biases in neural networks or *natural abstractions* (Chan et al., 2023) – concepts favored by the natural world that any cognitive system would naturally gravitate towards.

Hypothesis 4: Strong Universality

The *same* core features and circuits will universally and consistently arise across all neural network models trained on similar tasks and data distributions and using similar techniques, reflecting a set of fundamental computational *motifs* that neural networks inherently gravitate towards when learning.

Evidence for this hypothesis comes from *cross-species neural structures* in neuroscience, where similar neural structures and functions are found in different species (Kirchner, 2023). Additionally, machine learning models, including neural networks, tend to converge on similar features, representations, and classifications across different tasks and architectures (Chen et al., 2023a; Hacohen et al., 2020; Li et al., 2015; Bricken et al., 2023). Marchetti et al. (2023) provide mathematical support for emerging universal features.

While various studies support the universality hypothesis, questions remain about the extent of feature and circuit similarity across different models and tasks. In the context of mechanistic interpretability, this hypothesis has been investigated for neurons (Gurnee et al., 2024), group composition circuits (Chughtai et al., 2023), and modular task processing (Variengien & Winsor, 2023), with evidence for the weak but not the strong formulation (Chughtai et al., 2023).

3.4 Emergence of World Models and Simulated Agents

Internal World Models. World models are internal causal models of an environment formed within neural networks. Traditionally linked with reinforcement learning, these models are *explicitly* trained to develop a compressed spatial and temporal representation of the training environment, enhancing downstream task performance and sample efficiency through training on internal hallucinations (Ha & Schmidhuber, 2018). However, in the context of our survey, our focus shifts to *internal world models* that potentially form *implicitly* as a by-product of the training process, especially in LLMs trained on next-token prediction – also called GPT.

LLMs are sometimes characterized as *stochastic parrots* (Bender et al., 2021). This label stems from their fundamental operational mechanism of predicting the next word in a sequence, which is seen as relying heavily on memorization. From this viewpoint, LLMs are thought to form complex correlations based on observational data but cannot develop causal models of the world due to their lack of access to interventional data (Pearl, 2009).

An alternative perspective on LLMs comes from the *active inference* framework (Salvatori et al., 2023), a theory rooted in cognitive science and neuroscience. Active inference postulates that the objective of minimizing prediction error, given enough representative capacity, is adequate for a learning system to develop complex world representations, behaviors, and abstractions. Since language inherently mirrors the world, these models could implicitly construct linguistic and broader world models (Kulveit et al., 2023).

The *simulation* hypothesis suggests that models designed for prediction, such as LLMs, will eventually simulate the causal processes underlying data creation. Seen as an extension of their drive for efficient compression, this hypothesis implies that adequately trained models like GPT could develop *internal world models* as a natural outcome of their predictive training (janus, 2022; Shanahan et al., 2023).

Hypothesis 5: Simulation

A model whose objective is text prediction will simulate the causal processes underlying the text creation if optimized sufficiently strongly (janus, 2022).

In addition to theoretical considerations for emergent causal world models (Richens & Everitt, 2024; Nichani et al., 2024), mechanistic interpretability is starting to provide empirical evidence on the types of internal world models that may emerge in LLMs. The ability to internally represent the board state in games like chess (Karvonen, 2024) or Othello (Li et al., 2023a; Nanda et al., 2023b), create linear abstractions of spatial and temporal data (Gurnee & Tegmark, 2024), and structure complex representations of mazes, demonstrating an understanding of maze topology and pathways (Ivanitskiy et al., 2023) highlight the growing abstraction capabilities of LLMs. Li et al. (2021) identified contextual word representations that function as models of entities and situations evolving throughout a discourse, akin to linguistic models of dynamic semantics. Patel & Pavlick (2022) demonstrated that LLMs can map conceptual domains (e.g., direction, color) to grounded world representations given a few examples, suggesting they learn rich conceptual spaces (Gardenfors, 2004) reflective of the non-linguistic world.

The *prediction orthogonality* hypothesis further expands on this idea: It posits that prediction-focused models like GPT may simulate agents with various objectives and levels of optimality. In this context, GPT are simulators, simulating entities known as *simulacra* that can be either agentic or non-agentic, with different objectives from the simulator itself (janus, 2022; Shanahan et al., 2023). The implications of the simulation and prediction orthogonality hypotheses for AI safety and alignment are discussed in Section 6.

Hypothesis 6: Prediction Orthogonality

A model whose objective is prediction can simulate agents who optimize toward any objectives with any degree of optimality (janus, 2022).

In conclusion, the evolution of LLMs from simple predictive models to entities potentially possessing complex *internal world models*, as suggested by the *simulation* hypothesis and supported by mechanistic interpretability studies, represents a significant shift in our understanding of these systems. This evolution challenges us to reconsider LLMs' capabilities and future trajectories in the broader landscape of AI development.

4 Core Methods

Mechanistic interpretability (MI) employs various tools, from observational analysis to causal interventions. This section provides a comprehensive overview of these methods, beginning with a taxonomy that categorizes approaches based on their key characteristics (Section 4.1). We then survey observational (Section 4.2), followed by interventional techniques (Section 4.3). Finally, we study their synergistic interplay (Section 4.4). Figure 7 offers a visual summary of the methods and techniques unique to mechanistic interpretability.

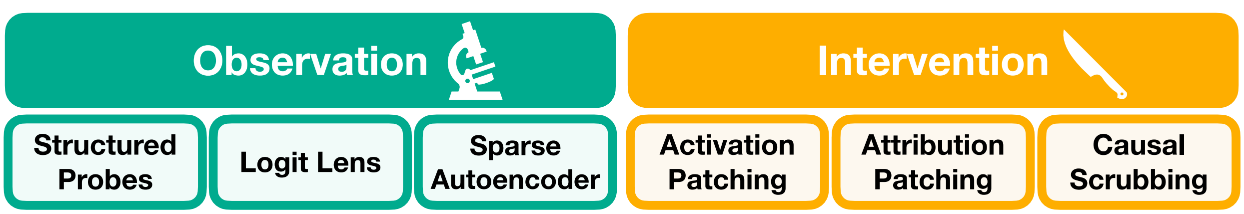

Figure 7: Overview of key methods and techniques in mechanistic interpretability research. Observational approaches include structured probes, logit lens variants, and sparse autoencoders (SAEs). Interventional methods, focusing on causal understanding, encompass activation patching variants for uncovering causal mechanisms and causal scrubbing for hypothesis evaluation.

4.1 Taxonomy of Mechanistic Interpretability Methods

We propose a taxonomy based on four key dimensions: causal nature, learning phase, locality, and comprehensiveness (Table 1).

The causal nature of methods ranges from purely observational, which analyze existing representations without direct manipulation, to interventional approaches that actively perturb model components to establish causal relationships. The learning phase dimension distinguishes between post-hoc techniques applied to trained models and intrinsic methods that enhance interpretability during the training process itself.